Discovery of Disease Relationships via Transcriptomic Signature Analysis Powered by Agentic AI

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INTRODUCTION

Modern disease classification relies largely on clinical symptoms and anatomical features, but this symptom-centered view often hides deeper molecular connections. Transcriptomic signatures, which reflect underlying cellular mechanisms, provide a biologically grounded lens to reexamine disease relationships. Previous studies have shown that gene expression profiles can uncover shared molecular programs across clinically distinct diseases, with implications for reclassification, biomarker discovery, and drug repurposing. However, scaling transcriptomic analysis across many diseases is challenging due to extensive preprocessing and variability across datasets.

To address this gap, we employ GenoMAS, an agentic AI system, to analyze 1,384 disease–condition pairs from the GenoTEX benchmark. This automated framework identifies transcriptomic signatures and constructs similarity networks that reveal both known and novel disease links. By extending the analysis to pathways, we highlight molecular mechanisms that transcend traditional taxonomy, offering new hypotheses for comorbidity, modulators such as obesity and hypertension, and drug repurposing opportunities for rare diseases.

METHOD

Data Collection:
We collected 1,384 disease–condition pairs from the GenoTEX benchmark, covering 132 diseases, 911 cohorts, more than 150,000 samples, and about 18,000 genes per cohort. Each pair combines a disease, such as Liver Cancer, with a biological or demographic condition, such as Obesity.

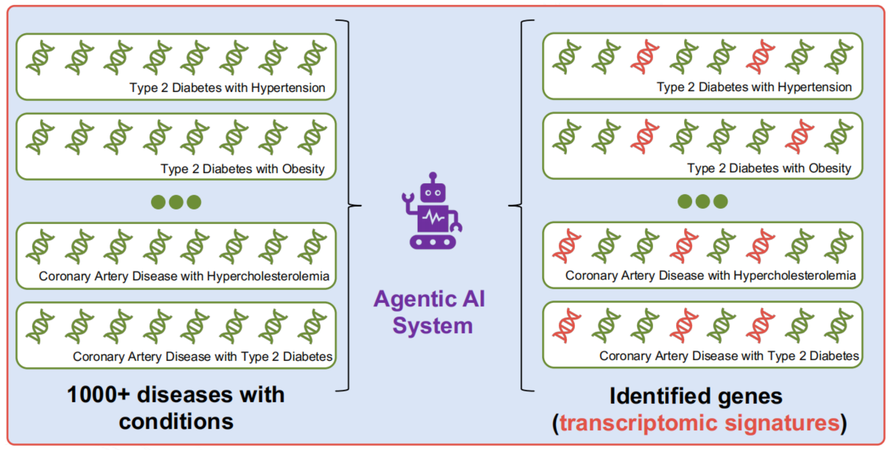
Transcriptomic Analysis with GenoMAS:
We applied GenoMAS, a multi-agent agentic AI system, to conduct end-to-end transcriptomic analysis. The system cleaned the data, applied Lasso regression ($|\beta| > 0.05$), and generated transcriptomic signatures for each pair.

Gene-Level Similarity Network:
To examine gene-level similarity, we tested overlaps between disease–condition pairs with a bidirectional hypergeometric test. We kept significant overlaps (FDR $\leqslant$ 0.05) and built a network where nodes represent the pairs and edge weights is the significance.

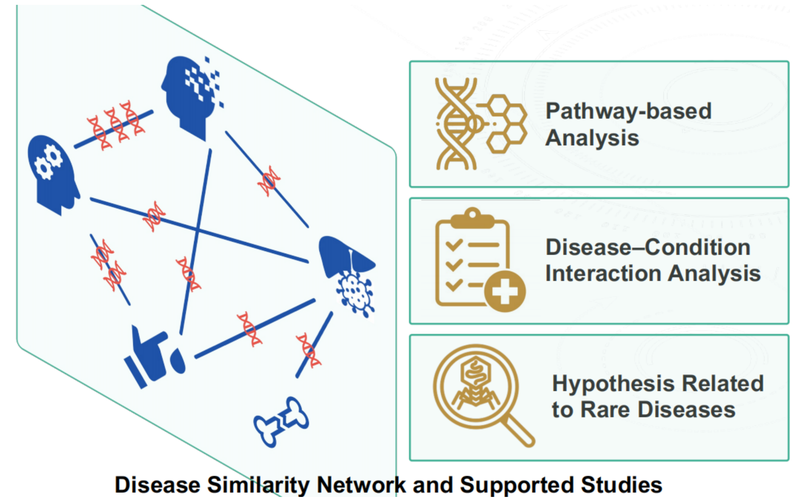
Pathway-Level Similarity Network:
We mapped significant genes to six annotation databases (GO, KEGG, HPO, etc), identified shared enriched pathways, and calculated a joint similarity score to measure functional convergence:

$$\text{Similarity Score} = \sum_{k=1}^{N} [log(1 - p_{1k}) + log(1 - p_{2k})]$$

Using these scores, we built a weighted pathway network (figure in the lower right) where node colors show ICD-10-CM categories, node size indicates connectivity, and edge width reflects significance.



Agentic AI System to Compute Transcriptomic Signatures



Disease Similarity Network and Supported Studies

ANALYSIS AND RESULTS

Gene-Level Similarity Reveals Hidden Cross-Category Links
Analyzing 1,384 disease–condition pairs, we identified ~65,000 significant overlaps. The network largely recapitulates ICD-10-CM categories but also revealed cross-category convergence. For instance, some cancers and immune disorders showed stronger similarity with musculoskeletal diseases than within their own categories.

Transcriptomic Similarity (Pathway-Level) Reflects Symptom-Based Taxonomy
Many high-scoring pairs matched known comorbidities. Canavan disease and Epilepsy shared pathways in sensory perception and GPCR signaling, while Bladder cancer and Endometrioid cancer overlapped in hormonal/tissue functions. These findings confirm that pathway-level similarity aligns with clinical taxonomy.

Cross-Category Similarity with Empirical Support
Biologically plausible links also emerged across categories. Hemochromatosis and Liver cancer overlapped in amino acid metabolism, mitochondrial function, and glycoprotein processing. Iron overload and oxidative stress provide a mechanistic bridge, consistent with epidemiological evidence.

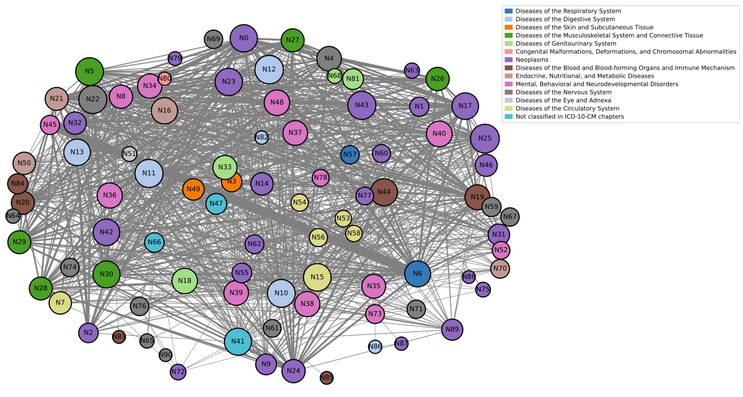
Unexpected Cross-Category Similarities
We also observed novel cross-category connections that have not been clinically reported but may suggest hypothesis-generating insights. For example, Gaucher disease and Kidney chromophobe share expression of A1BG, A4GNT, and A2M, converging on pathways in immune regulation, ECM remodeling, and glycosylation. Similarly, Alzheimer's disease and Ocular melanoma both show enrichment in lysine/tryptophan metabolism, pointing to a metabolic axis potentially involved in both neurodegeneration and tumor proliferation. These speculative results highlight overlooked molecular intersections that warrant further investigation.

Background Conditions Amplify Disease Comorbidity
Physiological factors can amplify similarity. Celiac disease and Uterine carcinosarcoma in obesity shared genes linked to bile secretion and amino acid metabolism. Hypertension was even more pervasive: 11 of the top 20 strongest overlaps involved hypertensive conditions, underscoring its role as a systemic amplifier of comorbidity.

Rare Diseases Share Molecular Programs with Common Conditions
Rare diseases also converged with well-characterized disorders. Autism spectrum disorder (ASD) aligned with Osteoporosis via lipid metabolism and with Type 1 diabetes via fatty acid oxidation and immune dysregulation. These overlaps suggest repurposing opportunities, including lipid-modulating agents and anti-inflammatory or metabolic therapies.

CONCLUSION

This study presents a transcriptomics-driven framework for rethinking disease relationships beyond traditional clinical boundaries. We propose a new similarity framework based on gene and pathway-level convergence, enabled by large-scale analysis of 1,300+ disease–condition pairs using the GenoMAS agentic AI system.

Our multi-layered analysis—from gene-level overlap to pathway-based convergence—captures both known comorbidities and novel cross-category links. We propose plausible molecular links across diseases, inferred from overlapping pathways related to metabolism, immune response, and cellular stress. Systemic conditions like obesity and hypertension emerge as modulators of transcriptomic similarity, while rare diseases such as autism spectrum disorder may benefit from therapeutic insights derived from better-characterized conditions.

By publicly sharing our full results and network resources, we hope to support hypothesis generation, validation, and translational research. More broadly, this work demonstrates how agentic AI can act as a biologically informed collaborator—scaling analysis while enabling mechanistic interpretation across complex disease landscapes.